AI CAREER FOR WOMEN

Commerce

Sales Business Intelligence Dashboard

A Project Report
submitted in partial fulfillment of the requirements
of
AICW project.

by

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Under the Guidance of
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This project focuses on analyzing a Retail Sales dataset to generate meaningful business insights and improve decision-making. The main problem identified was that the raw dataset contained inconsistencies such as duplicate records, missing values, and unstructured data, which made analysis difficult and reduced data reliability. The objective of this project was to clean and prepare the dataset, perform detailed analysis, and present the findings through interactive dashboards.

Microsoft Excel was used for data cleaning and preprocessing tasks such as removing duplicates, handling missing values, and organizing the dataset into a structured format. Power BI was used to create interactive dashboards and visualizations that display important business metrics including sales trends, regional performance, customer behaviour, and product category analysis. Different charts, filters, and KPIs were designed to make the dashboard user-friendly and easy to interpret.

The project resulted in clear visualization of sales performance and provided better understanding of business patterns and customer trends. The dashboards helped identify high-performing products, profitable regions, and sales growth opportunities. The analysis also improved the accessibility and interpretation of business data for effective reporting.

Overall, this project demonstrates how data cleaning, data analysis, and visualization techniques can be combined to transform raw retail data into meaningful insights that support better business decision-making and performance evaluation.

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CHAPTER 1

Introduction

1.1 Problem Statement:

Organizations in the retail sector generate large volumes of sales data on a daily basis. However, this data is often stored in raw form and contains issues such as duplicate records, missing values, and inconsistent formatting. This creates a major challenge in analyzing the data effectively and extracting meaningful insights.

This problem is significant because inaccurate or unclean data can lead to incorrect analysis, which directly affects business decisions, sales strategies, and overall profitability. Without proper data processing, organizations may fail to identify important trends, customer preferences, and high-performing products.

The primary users affected by this problem include business analysts, managers, and decision-makers who rely on accurate data to monitor performance and plan strategies. Retail companies depend heavily on data-driven insights to remain competitive in the market.

Existing approaches often involve manual data handling or basic tools that are time-consuming and prone to errors. Additionally, many systems lack interactive visualization and user-friendly dashboards for exploring data effectively.

1.2 Motivation:

The motivation behind choosing this project comes from the increasing importance of data analysis in today’s business environment and my interest in learning practical skills in this field. As a student of accounts, finance, and insurance, gaining knowledge about how data can be analyzed and used for decision-making is highly valuable for my future career.

This project is professionally meaningful as it allowed me to work with a real-world sample dataset obtained from Kaggle, which simulates actual retail sales data. It provided hands-on experience in data cleaning, analysis, and visualization using industry-relevant tools.

The project has practical applications in the retail sector, where businesses rely on sales data to track performance, identify trends, and improve strategies. Even though the dataset is a sample, the techniques used in this project can be applied to real business data.

The impact of this project lies in demonstrating how raw data can be transformed into meaningful insights through proper cleaning, analysis, and visualization. It also highlights the importance of business intelligence tools in improving reporting and supporting effective decision-making.

This project aligns with modern industry trends where organizations are increasingly using data analytics and business intelligence tools like Power BI to make informed decisions and gain a competitive advantage.

1.3 Objectives:

- To collect and understand the structure of the retail sales dataset obtained from Kaggle.
- To clean and preprocess raw data using Microsoft Excel by removing duplicates, handling missing values, and correcting data formats.
- To perform data analysis using Excel and Power BI to identify key patterns, trends, and insights in the dataset.
- To build interactive dashboards in Power BI that effectively present sales performance, product analysis, and regional insights.
- To enhance decision-making by transforming raw data into meaningful and actionable insights through data visualization techniques.

1.4 Scope of the project:

- This project covers data from a retail sales dataset obtained from Kaggle, which includes information related to sales transactions, products, and regional performance.
- The tools used in this project are limited to Microsoft Excel for data cleaning and preprocessing and Power BI for data visualization and dashboard creation.
- The analysis mainly focuses on data cleaning, data transformation, visualization, and generating insights through interactive dashboards.
- The project does not include real-time data integration, advanced predictive analytics, or machine learning models. It is limited to descriptive analysis and dashboard-based reporting.

CHAPTER 2

Literature Survey

2.1 Review relevant literature or previous work in this domain.

Data analysis and business intelligence have become essential in the retail industry for improving decision-making and business performance. Many organizations use data analytics techniques to analyze sales performance, customer behavior, and market trends. Previous studies and projects in this domain mainly focus on data cleaning, sales analysis, and dashboard development using tools such as Microsoft Excel and Power BI.

Research and industry practices show that raw retail datasets often contain duplicate records, missing values, and inconsistent formatting, which reduce data accuracy and make analysis difficult. Several existing retail analysis projects use business intelligence tools to transform raw data into meaningful visual reports and dashboards. Interactive dashboards are widely used for monitoring sales trends, product performance, regional analysis, and customer insights.

Many organizations also rely on visualization tools to simplify complex datasets and improve reporting efficiency. Power BI is commonly used because of its ability to create dynamic dashboards, charts, KPIs, and filters that help users understand business performance more effectively.

2.2 Mention any existing models, techniques, or methodologies related to the problem.

The existing methodologies used in retail sales analysis generally follow a structured process that includes:

1. Data Collection – Gathering retail sales data from databases, spreadsheets, or online sources.
2. Data Cleaning and Preprocessing – Removing duplicate records, handling missing values, correcting inconsistent formats, and organizing the dataset.
3. Data Analysis – Identifying trends, patterns, and relationships within the data using analytical techniques.
4. Data Visualization – Presenting findings through charts, graphs, KPIs, and dashboards for easier understanding.
5. Business Reporting – Generating reports that support business decisions and performance evaluation.

Microsoft Excel is commonly used for preprocessing and basic analysis, while Power BI is widely used for creating interactive dashboards and visual reports. These tools help organizations improve reporting accuracy and gain better insights from retail data.

2.3 Highlight the gaps or limitations in existing solutions and how your project will address them.

Many existing solutions rely heavily on manual data handling, which can be time-consuming and prone to errors. In several cases, datasets are not properly cleaned before analysis, resulting in inaccurate insights and poor decision-making. Some systems also provide only static reports that lack interactivity and flexibility for users.

Another limitation is that basic reporting tools may not effectively present large volumes of retail data in a clear and user-friendly manner. Users often face difficulties in identifying trends, comparing regional performance, and understanding product-wise sales patterns from raw datasets.

This project addresses these limitations by applying proper data cleaning and preprocessing techniques using Microsoft Excel and developing interactive dashboards using Power BI. The dashboards provide a clear and visually appealing representation of sales trends, product performance, and regional analysis. Interactive features such as filters, slicers, and KPIs improve user experience and make data interpretation easier.

By transforming raw retail data into structured visual insights, the project supports better business understanding and more effective decision-making.

CHAPTER 3

Proposed Methodology

3.1 System Design

3.2 The system design of this project follows a structured approach where raw retail sales data is processed step-by-step to generate meaningful insights. The workflow of the system moves from data collection and cleaning to analysis and dashboard visualization.

3.2.1 Registration:

In this stage, the retail sales dataset was collected from Kaggle in CSV/Excel format. The dataset contains transaction details such as product information, sales amount, customer segment, category, and regional data, which serve as the primary input for analysis. The dataset was then imported into Microsoft Excel and Power BI for further processing.

3.2.2 Recognition:

After importing the dataset, data preprocessing and analysis were carried out. Microsoft Excel was used for removing duplicate records, handling missing values, correcting inconsistent formats, and organizing the dataset into a structured form. Power BI was used to analyze the cleaned data and create interactive dashboards to identify sales trends, regional performance, customer behavior, and product-wise insights.

3.3 Modules Used

3.3.1 No Of Modules Used

Module 1 — Data Collection:
Retail sales data was obtained from Kaggle and loaded into Microsoft Excel and Power BI for analysis.

Module 2 — Data Cleaning:
The dataset was cleaned using Microsoft Excel by removing duplicate records, handling missing values, and correcting inconsistent data formats.

Module 3 — Data Analysis:
The cleaned dataset was analyzed using Excel functions such as sorting, filtering, pivot tables, and calculations. Power BI was used to identify sales patterns, trends, and key performance indicators.

Module 4 — Visualization :
Interactive dashboards were created in Power BI to present insights related to sales performance, product categories, regional analysis, and customer trends in a clear and user-friendly format.

Module 5 — Insight Generation:
The final stage involved summarizing important findings from the analysis to support better decision-making and business understanding.

3.4 Data Flow Diagram

A Data Flow Diagram (DFD) is a graphical representation of the "flow" of data through an information system, modeling its process aspects. A DFD is often used as a preliminary step to create an overview of the system, which can later be elaborated. DFDs can also be used for the visualization of data processing (structured design).

3.3.1. DFD Level 0

Input	Process	Output
Retail Sales Dataset	Retail Sales Analysis System	Power BI Dashboards

DFD Level 1 - Student Face Registration Module:

Process Step	Description
Data Input	Retail Sales dataset obtained from kaggle
Data Loading	Dataset loaded into Excel and Power BI
Data Cleaning	Removing duplicates, handling missing values, correcting formats
Data Analysis	Using Excel and Power BI to identify trends and patterns
Visualization	Creating dashboards and reports in Power BI
Output	Final insights and reports generated

DFD Level 1 - Student Face Recognition Module:

Process step	Description
Sales Analysis	**Analyzing sales performance across different regions and categories**
Product Analysis	**Identifying high-performing and low-performing product**
Regional Analysis	**Comparing sales across regions and markets**
Dashboard Filtering	**Applying slicers and filters for interactive analysis**
Report Generation	**Generating visual summaries and KPI reports**

DFD Level 1 - Concentration Analysis Module:

Process Step	Description
Trend Identification	**Identifying monthly and yearly sales trends**
Customer Insights	**Understanding customer purchasing behavior**
KPI Monitoring	**Monitoring profit, sales, and quantity metrics**
Data Visualization	**Presenting insights through charts and graphs**
Final Output	**Business insights for decision-making**

Advantages

- Reduces manual effort by using Microsoft Excel for efficient data cleaning and preprocessing.
- Provides interactive and visually appealing dashboards using Power BI.
- Helps users easily understand business performance through charts, KPIs, and reports.

- Improves decision-making by converting raw retail data into meaningful insights.
- Scalable approach that can be applied to larger datasets and different business domains.
  Uses widely available tools such as Excel and Power BI, making the project cost-effective and easy to implement.

3.5 Requirement Specification

3.5.1. Hardware Requirements:

Component	Specification
Processor	Intel Core i3
RAM	Minimum 4GB RAM
Storage	Minimum 50GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for dataset download and Power BI updates

Software Requirements:

Software	Details
Operating system	Windows 11
Microsoft Excel	Excel 2019
Power BI desktop	Latest Version
Web Browser	Google Chrome / Microsoft Edge

CHAPTER 4

Implementation and Result

4.1 Results of Face Detection

In this project, the first stage involved collecting the retail sales dataset from Kaggle and importing it into Microsoft Excel and Power BI. The raw dataset contained issues such as duplicate records, missing values, and inconsistent formatting.

Data cleaning techniques were implemented using Microsoft Excel to improve the quality and accuracy of the dataset. Duplicate records were removed, missing values were handled, and incorrect data formats were corrected. After preprocessing, the dataset became more structured and suitable for analysis.

The implementation of the data cleaning process resulted in improved data consistency and reliability, which helped in generating accurate insights and reports.

4.2 Results of Face Recognition

After the dataset was cleaned and organized, data analysis was performed using Excel and Power BI. Various analytical techniques such as sorting, filtering, pivot tables, and KPI calculations were used to identify sales trends and business patterns.

Interactive dashboards were developed in Power BI to visualize:

- Sales performance across different regions
- Product category analysis
- Monthly and yearly sales trends
- Customer purchasing patterns
- Profit and revenue analysis

The dashboards provided a clear understanding of business performance and enabled easy interpretation of complex retail data through charts, graphs, KPIs, slicers, and filters.

The implementation successfully transformed raw retail data into meaningful visual insights that support business reporting and decision-making.

4.3 Result of Concentration Analysis:

The final analysis focused on identifying important business insights from the retail sales dataset. Through visualization and reporting, the project highlighted:

- High-performing products and categories
- Regions with maximum sales and profit
- Sales growth trends over time
- Customer purchasing behavior patterns
- Overall business performance indicators

The Power BI dashboards improved the accessibility and presentation of retail sales data, making it easier for users to analyze performance and identify trends.

Overall, the project demonstrated how data cleaning, analysis, and visualization techniques can be effectively combined to convert raw retail sales data into actionable business insights.

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings:

The project successfully demonstrated how raw retail sales data can be transformed into meaningful business insights through data cleaning, analysis, and visualization techniques.

The key findings of the project include:

- Identification of sales trends across different time periods.
- Analysis of high-performing and low-performing product categories.
- Regional sales comparison to identify profitable markets.
- Improved understanding of customer purchasing patterns.
- Creation of interactive Power BI dashboards for effective reporting and visualization.
- Better data accuracy after removing duplicates and handling missing values using Microsoft Excel.

The dashboards helped in presenting complex retail data in a simple and user-friendly manner, which supports better business understanding and decision-making.

5.2 Git Hub Link of the Project:

https://github.com/Dhanalakshmi-0417/Sales-Business-Intelligence-Dashboard

5.3 Video Recording of Project Demonstration:

https://drive.google.com/file/d/1pBv-xkNmtMtiNDAbJEYAVBcE-b6ZkvyF/view?usp=sharing

5.4 Limitations: Discuss the limitations of the current model or approach.

Although the project achieved its objectives successfully, there are certain limitations:

- The dataset used in the project is a sample dataset obtained from Kaggle and may not represent real-time business data.
- The project is limited to descriptive analysis and does not include predictive analytics or forecasting models.
- Real-time data integration was not implemented.
- The dashboards depend on the quality and completeness of the dataset provided.
- Advanced business intelligence and automation features were not included in the project.

5.5 Future Work:

The project can be improved further in the following ways:

- Integration of real-time retail sales data for live dashboard updates.
- Implementation of predictive analytics and sales forecasting techniques.
- Addition of more advanced Power BI features and automated reporting systems.

- Expansion of the dataset to include customer feedback and inventory details for deeper analysis.
- Development of more detailed business performance metrics and comparative analysis dashboards.

5.6 Conclusion:

This project focused on analyzing a retail sales dataset using Microsoft Excel and Power BI to generate meaningful business insights. The project successfully demonstrated the importance of data cleaning, preprocessing, analysis, and visualization in transforming raw data into useful information.

The interactive dashboards created in Power BI helped in understanding sales trends, product performance, and regional analysis effectively. The project also enhanced practical knowledge in data analytics and business intelligence tools.

Overall, the project highlights how data-driven analysis can support better business decision-making, improve reporting efficiency, and provide valuable insights for the retail industry.

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[5] Research Paper Link: https://journal.ijresm.com/index.php/ijresm/article/view/2670?utm_source=chatgpt.com

Appendices

Figure 1: Image of Executive Overview from the dashboard

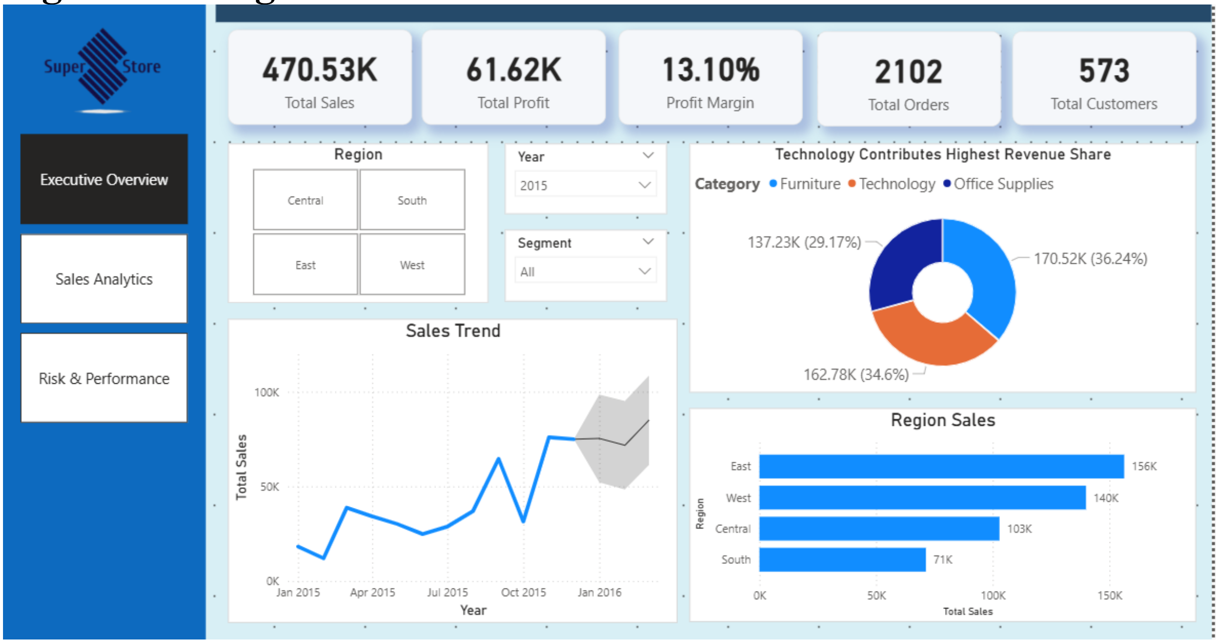

Figure 2: Image of Sales Analytics page from the dashboard

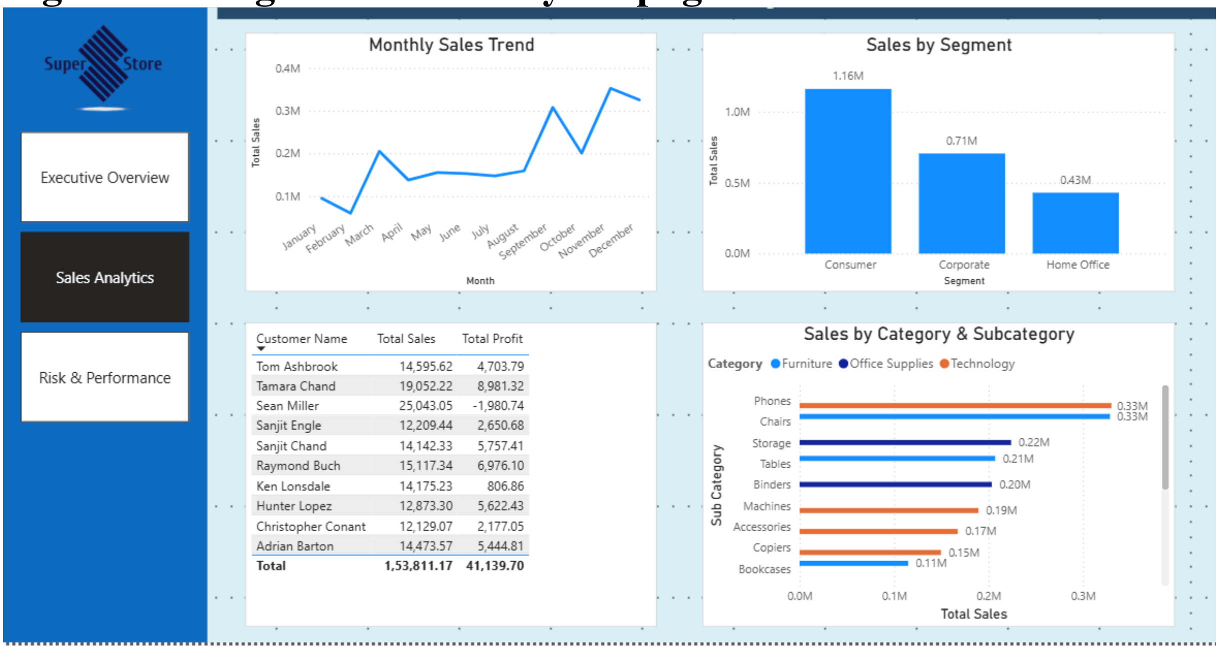

Figure 3: Image of risk & performance page from the dashboard

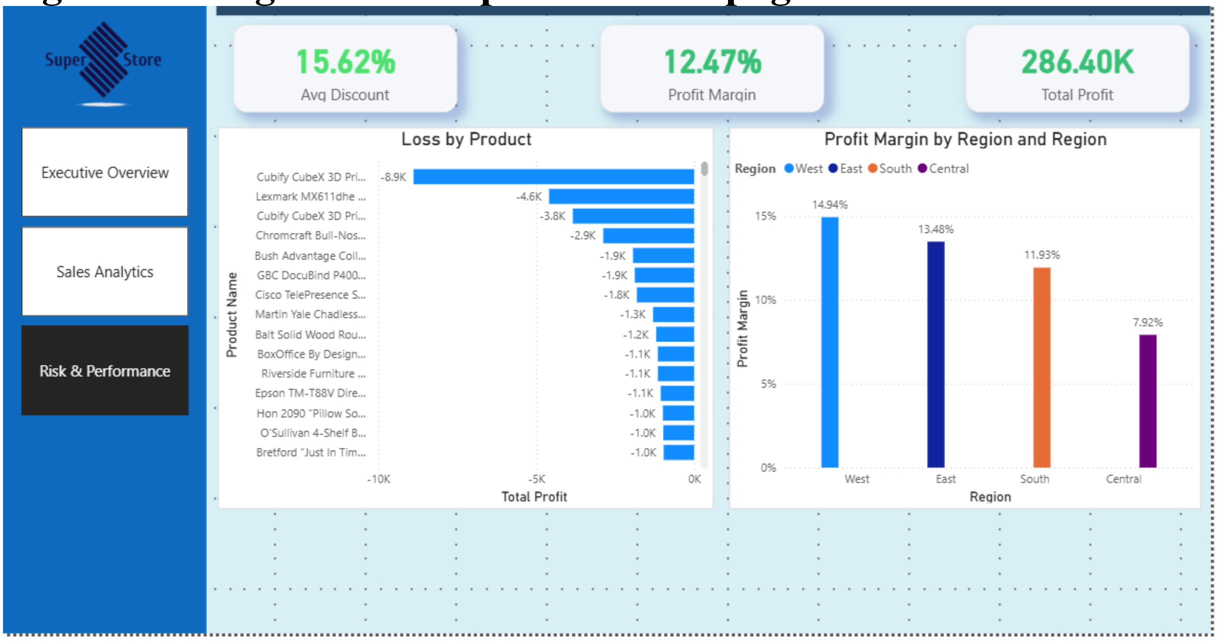

Figure 4: Image of tooltip used from the dashboard